IAM HOME LAB PORTFOLIO

Lab 01

Active Directory Environment Setup & On-Premises Identity Operations

Opeyemi Ogundimu • IAM Analyst | Identity & Access Management
github.com/okpe027/ https://github.com/okpe027/IAM-HomeLab-Portfolio

Lab	Lab 01 of 20 — Phase 1 (Active Directory & On-Prem Identity)
Completed	April 15, 2026
Environment	VMware Workstation • Windows Server 2022 • Active Directory
What I built	A working Domain Controller for FictionalCorp.local, a full OU structure, role-based security groups, and two provisioned user accounts
Domain	FictionalCorp.local | Server: DC01 | IP: 192.168.10.10
Users	Marcus Williams (permanent employee) and Jessica Park (90-day contractor)
Foundation	This lab is the on-prem foundation that every subsequent lab builds on

1. What This Lab Is About

Every IAM lab program starts here. Before you can talk about Conditional Access policies or Okta integrations or Zero Trust, you need to understand where identity management came from — and for most organizations, that means Active Directory.

This lab builds the on-premises foundation for FictionalCorp, a fictional company I'm using as my lab environment throughout this entire program. What I set up here — the domain, the OU structure, the security groups, the naming conventions — will carry through all 20 labs. In Lab 05, I'll sync this entire environment to Microsoft Entra ID using Azure AD Connect, creating a proper hybrid identity setup that mirrors what you find in most real enterprises.

But before any of that, there's a more immediate point to this lab: learning to think like an IAM analyst from day one. That means working from tickets, documenting every action, and making deliberate decisions about structure rather than just clicking through wizard screens.

By the end of this lab I had:

- A fully functioning Windows Server 2022 Domain Controller on VMware
- A clean OU structure designed around real department logic
- Role-based security groups that match how access should actually work
- Two provisioned user accounts processed from mock HR tickets
- A provisioning log that serves as a proper audit trail

2. The Lab Environment

I'm running this on VMware Workstation on my personal machine. The VM is a Windows Server 2022 Standard Evaluation installation — the Desktop Experience version, because the GUI is essential for a learning environment.

One thing I set up that's worth noting: the VM has two network adapters. A NAT adapter handles internet connectivity (which I'll need in Lab 05 for Azure AD Connect downloads), and a Host-only adapter keeps the AD traffic on a private 192.168.10.x network. This separation is closer to how production environments work — domain controllers are typically on isolated management networks, not directly internet-connected.

Setting	Value	Why
Server name	DC01	Short, clean, matches industry convention for Domain Controller 01
Domain	FictionalCorp.local	Internal domain suffix — .local is standard for non-public AD domains
IP address	192.168.10.10	Static IP — DC must never change its IP or AD breaks
DNS (preferred)	127.0.0.1	Server points to itself for DNS after AD promotion
Network (NAT)	Internet access	Needed for Azure AD Connect download in Lab 05
Network (Host-only)	192.168.10.x	Private AD network — isolated from home network
RAM / CPU	4 GB / 2 vCPU	Minimum comfortable spec for AD + DNS services

3. The OU Structure I Designed

This was my first real design decision in the lab. Organizational Units in Active Directory are folders that organize accounts by department, and where you place a user determines which Group Policies apply to them and how easy they are to find during an audit.

I didn't just create one OU per department. I also added two OUs that are easy to overlook but are operationally important:

WHY DISABLED_USERS AND CONTRACTORS GET THEIR OWN OUS

Disabled_Users: When someone is terminated, you never delete their account immediately. You disable it and move it here for a retention period (30 days standard, 90 days for legal holds). Having a dedicated OU makes it instantly clear during any audit which accounts are inactive and why they still exist.

Contractors: Mixing contractor accounts into department OUs creates confusion during access reviews. A dedicated Contractors OU makes them immediately visible, allows specific Group Policies to apply to them only, and when these accounts sync to Entra ID in Lab 05, the OU path confirms the account type without needing any additional attributes.

OU Name	Who goes here	Notes
HR	All HR permanent staff	Linda Okafor, Priya Sharma (provisioned in Lab 02)
Finance	All Finance permanent staff	Alex Chen, Rachel Thompson, Michael Obi (Lab 02)
IT	All IT permanent staff	David Osei, Karen Nwosu (Lab 02)
Sales	All Sales permanent staff	Marcus Williams (this lab), James Adeyemi, Fatima Al-Hassan (Lab 02)
Disabled_Users	All terminated or inactive accounts	Where offboarded accounts live until deleted — used in Lab 03
Contractors	All time-limited contract workers	Jessica Park (this lab) — separate from permanent staff

4. Security Groups

I created the foundational security groups in this lab and expanded them significantly in Lab 02 when I built the full RBAC role matrix. The key principle I applied here is one that's worth stating clearly because it comes up in every IAM interview:

NEVER ASSIGN PERMISSIONS TO INDIVIDUAL USERS

Permissions go on groups, users go into groups. Always.

If you assign access directly to individual user accounts, you create an unmanageable mess. When that person changes roles, you have to audit and update every individual permission. With groups, a role change is a single group membership update that immediately adjusts all associated permissions across every system the group has access to.

This is RBAC in practice. It's also the first thing an auditor checks.

The initial groups I created in this lab:

Group name	OU	Purpose
GRP-HR-Users	HR	Standard access for HR department staff
GRP-Finance-Users	Finance	Standard access for Finance department staff
GRP-Finance-InvoiceCreate	Finance	Invoice creation permission — SoD control (used in Lab 02 and Lab 14)
GRP-Finance-InvoiceApprove	Finance	Invoice approval permission — SoD control (used in Lab 02 and Lab 14)
GRP-IT-Users	IT	Standard access for IT department staff
GRP-Sales-Users	Sales	Standard access for Sales department staff

The two Finance SoD groups might seem premature at this point, but I added them deliberately. In Lab 02 I'll demonstrate a Segregation of Duties violation where a single user holds both Invoice Create and Invoice Approve rights simultaneously. Setting these groups up now means the structure is ready when that scenario runs.

5. The Provisioning Tickets

I processed two accounts in this lab. One permanent employee and one contractor. The difference between how they're handled is more significant than it might first appear.

Before creating either account, I set up a provisioning log spreadsheet. Every account action gets logged there — who was created, when, what groups they were assigned to, any expiry dates, and the ticket number that authorized the change. This is the audit trail that compliance teams ask for.

INC-0001 — Marcus Williams — New Sales Representative

```
TICKET #:     INC-0001
Date:         April 15, 2026
Requested by: Sarah Johnson (HR Manager)

New hire:     Marcus Williams
Job title:    Sales Representative
Department:   Sales
Account type: Permanent employee
Username:     m.williams
Access:       Standard Sales user access
Notes:        None
```

Standard provisioning. I created the account in the Sales OU, assigned it to GRP-Sales-Users, and set the password to force a change on first login. Nothing complicated here, but the discipline matters — correct OU, correct group, correct password policy, logged in the provisioning spreadsheet.

Field	Value
Username	m.williams
OU placement	FictionalCorp.local/Sales
Groups assigned	GRP-Sales-Users
Account expiry	None — permanent employee
Password policy	User must change password at next logon
Account type	Permanent

INC-0002 — Jessica Park — IT Support Contractor

```
TICKET #:       INC-0002
Date:           April 15, 2026
Requested by:   Tom Richards (IT Manager)

Contractor:     Jessica Park
Job title:      IT Support Contractor
Department:     IT
Account type:   Contractor — time-limited
Username:       j.park.contractor
Contract end:   July 14, 2026
Access:         IT support access only — no admin rights
Notes:          Account must expire on contract end date
```

This is where it gets more interesting. A few deliberate choices here worth explaining:

THE .CONTRACTOR SUFFIX

The username j.park.contractor — not just j.park — is a naming convention many organizations use. It makes contractor accounts immediately identifiable when browsing AD, running reports, or doing access reviews. You don't have to open the account to know what type it is. Small detail. Makes a real difference at scale.

THE ACCOUNT EXPIRY DATE

This is the most important setting on a contractor account. I set the expiry to 90 days from today. When that date arrives, AD automatically disables the account. No human has to

> remember. No ticket needs to be raised. No risk of access lingering past the contract end.
>
> Most IAM incidents involving contractors happen because someone forgot to offboard them. Account expiry eliminates that failure mode entirely. Set it at provisioning time, every time.

Field	Value
Username	j.park.contractor
OU placement	FictionalCorp.local/Contractors
Groups assigned	GRP-IT-Users
Account expiry	July 14, 2026 (90 days — set at provisioning)
Password policy	User must change password at next logon
Account type	Contractor

6. The Provisioning Log

Every account action in this lab was logged in a spreadsheet I'm calling the FictionalCorp Provisioning Log. It tracks date, ticket number, full name, username, department, OU placement, groups assigned, account expiry, and who made the change.

This log carries forward through all 20 labs. Every time I provision, modify, or terminate an account, I add a row. By Lab 20 it will be a complete record of every identity event in the FictionalCorp environment — which is exactly what an IAM team's ticketing system and audit log looks like in production.

Ticket	Name	Username	OU	Groups	Expiry	Type
INC-0001	Marcus Williams	m.williams	Sales	GRP-Sales-Users	None	Permanent
INC-0002	Jessica Park	j.park.contractor	Contractors	GRP-IT-Users	July 14, 2026	Contractor

7. Decisions I Made and Why

A lab isn't just about running through steps. It's about understanding why each step matters. These are the decisions I made in this lab that I'd want to be able to explain in an interview.

Why separate adapters for NAT and Host-only?

A single Host-only adapter isolates the VM with no internet access. Adding NAT alongside it gives me connectivity for downloading Azure AD Connect in Lab 05 and connecting to the Azure tenant. This is also closer to how production domain controllers are configured — on a segmented management network with controlled external access, not directly internet-facing.

Why set 127.0.0.1 as the DNS server before promoting to DC?

After AD promotion the server becomes its own DNS server. If I point it to an external DNS address first, DNS breaks when the AD role activates. Self-referencing DNS is the right setting for a domain controller and it needs to be in place before the promotion runs.

Why put contractors in their own OU instead of in IT?

Three reasons. First, it makes contractors immediately identifiable without opening accounts. Second, I can apply contractor-specific Group Policies at the OU level — stricter password policies, shorter lockout thresholds, no local admin. Third, when this environment syncs to Entra ID in Lab 05, the OU path appears as an attribute on the synced user, making it visible in the cloud too.

Why force a password change at first logon?

I set the temporary password. The user should own their own password from day one. Leaving the provisioning password in place means I technically know a user's credentials, which is a security violation. Force change at first logon eliminates that issue.

8. How I Would Talk About This in an Interview

WHEN THEY ASK: WALK ME THROUGH YOUR PROVISIONING PROCESS

I'd say something like this:

When a provisioning request comes in, I work from the ticket. I need the person's full name, job title, department, and whether they're a permanent employee or a contractor before I create anything. I place the account in the correct OU based on their department, assign them to the role-based security group for that role following least privilege — so they get exactly what their job function requires and nothing more.

For contractors specifically, I always set an account expiry matching the contract end date at provisioning time. That way the account disables automatically when the contract ends without anyone having to remember to do it. And the .contractor suffix in the username makes them instantly identifiable in any audit or access review.

Every action gets logged in our provisioning tracker with the ticket number, so there's a

> clean audit trail for every account that exists in the environment.

9. How This Connects to Azure

Everything I built in this lab — the domain, the OUs, the groups, the users — will sync to Microsoft Entra ID in Lab 05 when I configure Azure AD Connect. That sync is what creates the hybrid identity environment that most enterprises actually run.

The reason I'm mentioning it here is that the decisions I made in this lab have downstream consequences in the cloud. The OU structure appears as an attribute on synced users in Entra ID. The group names carry across. The naming conventions I established here become the naming conventions I'm working with in every cloud lab going forward.

> **BUILD IT CLEAN NOW**
> A messy on-premises structure creates a messy cloud structure.
> The time to get the naming conventions, OU design, and group structure right is Lab 01.
> Everything I set up here carries forward for all 20 labs.

10. Portfolio Screenshots

These are stored in the Lab01_Portfolio folder and uploaded to GitHub under /Lab01-AD-Environment/Screenshots/.

#	What the screenshot shows	Why I captured it
1	Full OU structure in ADUC	Shows the complete FictionalCorp.local domain tree with all 6 OUs
2	Marcus Williams — Member Of tab	Confirms GRP-Sales-Users assignment and least privilege
3	GRP-Sales-Users — Members tab	Shows Marcus as a member — proves group-based access assignment
4	Jessica Park — Account tab showing expiry date	Confirms 90-day contractor expiry is set correctly
5	Completed provisioning log spreadsheet	Shows both INC-0001 and INC-0002 entries as the audit trail

11. What I Produced

Deliverable	Where it lives	Status
Provisioning log (INC-0001, INC-0002)	/Docs/	**Done ✓**
Lab 01 screenshots (x5)	/Lab01-AD-Environment/Screenshots/	**Done ✓**
This documentation	/Lab01-AD-Environment/	**Done ✓**

12. What Comes Next

LAB 02 — RBAC DESIGN, ROLE ENGINEERING & ACCESS GOVERNANCE

Lab 02 is where I take everything set up here and turn it into a proper access control model. Before touching anything in AD, I build a formal role matrix in Excel — mapping every job title to its permitted systems and documenting the SoD policy.

Then I implement it in AD, simulate a privilege creep scenario where a user accumulates excess permissions through an incomplete role change, detect it through an access audit, and remediate it. The lab ends with a quarterly access certification — the kind of compliance exercise that IAM teams run every quarter in regulated industries.

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